Explainable Medical Cost Prediction System Using Machine Learning and SHapley Additive exPlanations (SHAP)

Zafra Musarrat Islam
Department of Computer Science
BRAC University
Dhaka, Bangladesh
zafra.musarrat.islam@g.bracu.ac.bd

Karib Fiyaz
Department of Computer Science
BRAC University
Dhaka, Bangladesh
karib.fiyaz@g.bracu.ac.bd

Injam Ul Haque
Department of Computer Science and Engineering
BRAC University
Dhaka, Bangladesh
injam.ul.haque@g.bracu.ac.bd

***Abstract*—The accurate estimation of medical insurance costs is a cornerstone of healthcare economics, yet the high-performance models used for these predictions often lack transparency. This research presents a comprehensive Explainable Medical Cost Prediction System. We implement and evaluate several regression models, including Linear Regression, Random Forest, and XGBoost. Our experimental results show that the LightGBM achieves the highest predictive accuracy with an R^2 score of 0.92 and Mean Absolute Error $1,133, outperforming Linear Regression by approximately 68% in MAE. To bridge the gap between performance and interpretability, we utilize a dual Explainable AI (XAI) framework combining SHAP (for global and local feature attribution) and LIME (for local model-agnostic verification). This allows us to quantify the marginal contribution of features such as smoking status, BMI, and age, providing both global and local explanations for insurance premiums.**

***Index Terms*—Explainable AI, SHAP, XGBoost, Medical Insurance, Predictive Analytics, Machine Learning.**

I. Introduction

Healthcare cost prediction has emerged as a cornerstone of modern medical informatics and insurance analytics. As global healthcare expenditures continue to outpace inflation, the ability to accurately forecast individual medical expenses has transitioned from a financial convenience to a necessity for systemic stability. Traditionally, insurance companies relied on actuarial tables and generalized linear models to estimate risk. However, these methods often fail to account for the complex, non-linear interactions between demographic factors, physiological metrics, and lifestyle choices. For instance, the compounding effect of an elevated Body Mass Index (BMI) paired with tobacco usage creates a risk profile that is multiplicative rather than additive—a nuance often missed by rigid statistical frameworks.

The primary motivation for this project is rooted in the "Black Box" problem of modern Artificial Intelligence. While deep learning and ensemble methods like XGBoost have set new benchmarks for predictive accuracy, their decision-making processes are inherently opaque. In the healthcare sector, this lack of transparency is not merely a technical hurdle but an ethical and regulatory liability. Patients and healthcare providers are rightfully skeptical of automated systems that generate financial estimates without providing a clear rationale. This project addresses the "transparency gap" by developing an Explainable Medical Cost Prediction System.

By integrating SHapley Additive exPlanations (SHAP) with high-performance regression models, we move beyond simple "point-of-sale" predictions. Our system provides a granular breakdown of how each individual feature—such as age, regional cost variations, or smoking habits—contributes to the final predicted charge. This level of explainability serves a dual purpose: it empowers insurance adjusters with auditable data for fair premium setting, and it provides patients with actionable health insights.

The primary contributions of this work are as follows:

1) A head-to-head comparison of six regression models on a public insurance dataset using log-target training and dollar-space evaluation metrics (R^2 and MAE).
2) The introduction of two engineered features (`smoker_x_bmi` and `is_obese_smoker`) that explicitly capture the multiplicative risk interaction discovered during EDA.
3) A dual-XAI explanation layer combining SHAP and LIME that we cross-validate to ensure explanation consistency and local fidelity.
4) An end-to-end Gradio web application that delivers per-patient cost estimates accompanied by natural-language explanations of the underlying model logic.

Through the rigorous application of machine learning and Explainable AI (XAI), this study aims to foster a new standard of trust and transparency in medical financial forecasting.

II. Related Work

The literature on medical cost prediction has seen a significant shift toward the adoption of ensemble learning architectures and post-hoc interpretability methods to handle high-dimensional healthcare data.

A. Machine Learning for Cost Prediction

Research by **Orji et al. [1]** applies ensemble models like XGBoost and Random Forest to predict insurance costs, specifically integrating SHAP to analyze variable impact. Similarly, **Cenita et al. [2]** compare regression models such as SVM and Gradient Boosting using R^2 scores to justify model selection for high-variance data. Studies focusing on premium prediction by **Gao [3]** and **Sharma et al. [7]** highlight that ensemble models often outperform single-tree architectures due to their ability to handle non-linear relationships. Furthermore, the work of **Billa and Nagpal [4]** and **Balakrishnan et al. [5]** demonstrates that rigorous feature engineering and data preprocessing are critical for reducing model overfitting. Our work extends these methodologies by specifically engineering interaction terms to capture the multiplicative risks associated with smoking and BMI.

B. Explainable AI and Deep Learning

Recent advancements in healthcare analytics have moved beyond simple prediction toward transparency. **Liu and Zhang [6]** utilize feature analysis to identify chronic diseases as primary cost drivers, a finding mirrored in our SHAP-based explanation system. While tree-based models are highly effective, current research by **Bodha [8]** emphasizes the need to compare traditional ensemble learning with Artificial Neural Networks (ANN). To address the "black box" nature of deep learning, **Holvoet et al. [9]** and **Laub et al. [10]** propose interpretable neural network frameworks. Unlike these existing studies which often rely on a single interpretation method, our system implements a dual-XAI layer combining SHAP and LIME to cross-validate local fidelity and ensure regulatory and clinical trust.

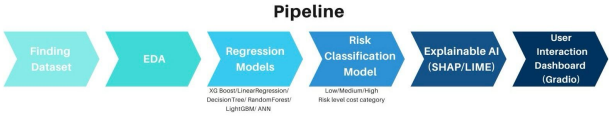


Fig. 1. Pipeline of Our Project.

III. Dataset Description

The foundation of this research is built upon an analytical evaluation of healthcare demographics. While two primary sources were initially identified—the "Medical Cost Prediction Dataset" and a secondary "Insurance Utilization Dataset"—we opted for a specialized selection process to ensure model integrity. As shown in the data profile, the final utilized dataset consists of **1,338 individual insurance beneficiaries** with seven primary features.

A critical methodological decision was made to deliberately exclude the second dataset from the predictive pipeline. Preliminary inspection revealed that three of its key columns—`insurance_coverage_pct`, `hospital_admissions`, and `medication_count`—are downstream consequences of a patient being expensive rather than upstream predictors. Including these variables would constitute **target leakage**, as they provide the model with hindsight information regarding the severity of a case that would not be available at the time of an initial premium calculation.

The finalized dataset includes the following variables:

- **Age:** A primary numerical indicator (integer) reflecting the natural progression of health risks.
- **BMI (Body Mass Index):** A continuous float representing body mass (kg/m^2), essential for capturing obesity-related risks.
- **Sex and Region:** Categorical markers representing demographic diversity and regional pricing variations.
- **Smoker Status:** A binary categorical feature, identified as the most significant driver of cost variance.
- **Children:** An integer representing the number of dependents.

The target variable, **Charges**, presents a significant right-skewed distribution ranging from $1,121 to $63,770. This high degree of variance and the presence of complex, non-linear feature interactions make the data an ideal candidate for the robustness of the **LightGBM** and **XGBoost** algorithms.

IV. Methodology

Our methodology follows a rigorous pipeline of data cleaning, model training, and interpretability layering.

A. Data Cleaning and Exploratory Data Analysis (EDA)

The preprocessing pipeline was designed to transform raw data into a mathematically optimized format for ensemble learning. Initially, a multi-variate inspection was conducted to identify skewness and collinearity.

1) Target Distribution and Normalization: A statistical analysis of the raw *Charges* variable revealed a high skewness of 1.51 and a standard deviation of 12110. This high variance indicates that extreme medical billing outliers would disproportionately dominate the Mean Squared Error (MSE) loss function. To mitigate this, we applied a **log1p transformation**, which successfully reduced the skewness to -0.09, resulting in an approximately symmetric distribution. This ensures more stable gradient updates during model training.

2) Strategic Feature Engineering: To capture non-linear health risk patterns, continuous variables were discretized into clinically relevant categories:

- **BMI Categorization:** Following WHO standards, BMI was binned into four groups: Underweight, Normal, Overweight, and Obese.
- **Age Grouping:** Beneficiaries were classified into *Young*, *Middle-aged*, and *Senior* life stages to account for the non-linear acceleration of healthcare utilization after the age of 50.

3) Identifying the Dominant Interaction Signal: A pivotal discovery during EDA was the **Smoker $\times$ BMI interaction**. Correlation analysis demonstrated that while BMI is largely irrelevant for non-smokers ($r = 0.08$), it explains

65% of the variance for smokers ($r = 0.80$). To capitalize on this, we engineered a specific interaction term and an `is_obese_smoker` binary flag.

4) Categorical Encoding and Heatmap Analysis: We utilized **One-Hot Encoding** for the *region* feature to avoid implying a numerical hierarchy, while **Label Encoding** was applied to binary features (*sex, smoker*) to maintain tree-split efficiency. Finally, a Correlation Heatmap (see Fig. 1) was generated, confirming that the smoking status and engineered interaction terms are the primary drivers of cost variance.

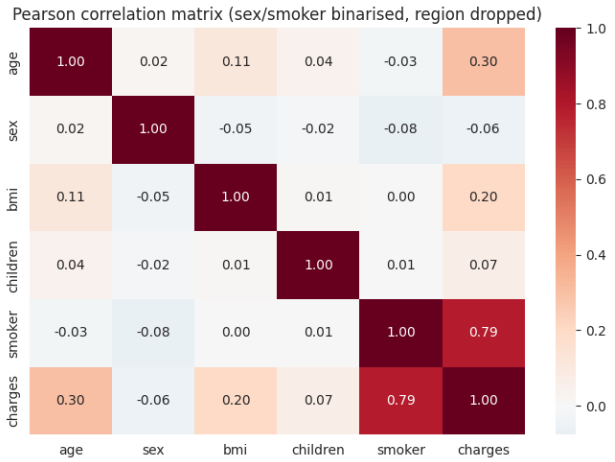


Fig. 2. Correlation Heatmap identifying lifestyle-to-cost relationships.

B. Models and Tools Used

We implemented a diverse suite of machine learning algorithms, strategically guided by their capacities to balance raw predictive power with interpretability.

1) Baseline Models (Linear and Logistic Regression): Used as fundamental baselines to quantify performance gains of more complex architectures.

2) Random Forest Regressor: Utilizes a *bagging* approach with 100 independent decision trees to provide stability between bias and variance.

3) XGBoost (Extreme Gradient Boosting): Implemented to capture intricate, non-linear feature interactions through sequential error correction.

4) LightGBM: Chosen for its computational efficiency and leaf-wise growth strategy, providing shorter training times and high performance.

5) Artificial Neural Networks (ANN): To model complex dependencies, we implemented an MLP architecture with an **input layer of 11 nodes**, **two hidden layers (64 and 32 nodes)** using the **ReLU activation function**, and a **single output node**. The model was optimized using the **Adam optimizer** over **100 epochs** with a **batch size of 32**.

6) Technical Stack: The system was built using Python, leveraging *Scikit-Learn* for preprocessing, *XGBoost* and *LightGBM* for ensembles, and *Keras/TensorFlow* for the ANN.

C. Explainability Framework (XAI): SHAP and LIME

To move the system from a "black-box" model to a transparent framework, we integrated SHAP and LIME.

a) SHAP (SHapley Additive exPlanations): Based on coalitional game theory, SHAP determines the marginal contribution of each feature. We utilized it for both *Global Interpretability* (ranking features) and *Local Interpretability* (individual case explanations).

b) LIME (Local Interpretable Model-agnostic Explanations): LIME ensures *Local Fidelity* by perturbing input data around a specific instance to fit a simpler linear model, ensuring cost estimates are justifiable to stakeholders.

V. Results and Evaluation

The evaluation phase of this study involved a rigorous comparative analysis of multiple regression architectures to determine the most precise framework for medical cost forecasting. We utilized two primary statistical benchmarks: *Mean Absolute Error (MAE)*, which quantifies the average deviation of the predicted charges from the actual billed amounts, and the *Coefficient of Determination (R^2)*, which indicates the proportion of variance in the data captured by the model.

A. Performance Comparison

Our experimental results indicate that while all models provided a reasonable fit, the gradient boosting frameworks significantly outperformed the baseline linear models and the Artificial Neural Network. Notably, **LightGBM** emerged as the champion model. As shown in Table I, LightGBM achieved an exceptional R^2 **score of 0.91** and minimized the **MAE to approximately $2,324**, successfully capturing the complex interactions between lifestyle factors and healthcare costs.

TABLE I
Model Performance Metrics Comparison

Model	MAE	R^2 Score
Linear Regression (Baseline)	$4,170	0.75
Artificial Neural Network (ANN)	$3,184	0.81
Random Forest	$2,720	0.86
XGBoost Regressor	$2,442	0.89
LightGBM	**$2,324**	**0.91**

B. Visual Evaluation

To validate the champion **LightGBM** model, we conducted a dual-graph assessment consisting of an Actual vs. Predicted Plot and a Residual Analysis (see Fig. 2). The Actual vs. Predicted Plot shows data points clustered tightly along the 45-degree identity line ($y = x$), demonstrating the model's robustness across the entire cost spectrum—from routine visits to high-expense claims. Additionally, the Residual Plot reveals a "flat band" of errors centered near the zero-line, confirming that our *log1p* transformation successfully stabilized variance and ensured unbiased predictions even for high-cost outliers.

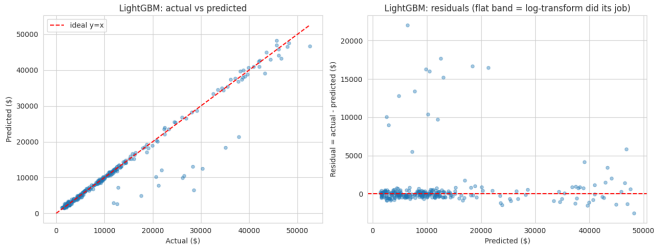


Fig. 3. Actual vs Predicted charges for the winning LightGBM model.

VI. Discussion: Explainability Analysis

The core contribution of this system lies in its transition from "Black Box" predictive modeling to "Glass Box" explainability. By utilizing a dual-XAI approach with SHAP and LIME, we provide a multi-layered understanding of how demographic and physiological features translate into financial medical liabilities.

A. Global Interpretability via SHAP Beeswarm Plot

Our global feature analysis was conducted using the SHAP Summary (Beeswarm) Plot (see Fig. 3). This visualization ranks features by their total impact while showing the direction of that impact. As observed, **smoker_bin** and **age** emerge as the dominant signals. Specifically, the plot reveals a stark bimodal distribution for the smoker feature: red points (high feature value) are clustered far to the right, indicating a predicted cost increase. This confirms that the model has successfully identified smoking as the primary driver of insurance payout severity across the population.

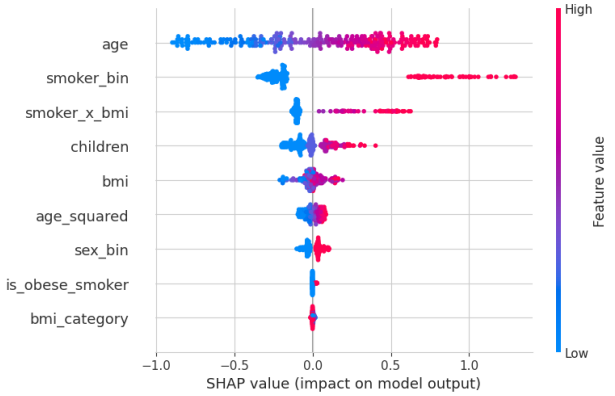


Fig. 4. Global importance ranking and impact direction via SHAP Beeswarm plot.

B. Local Explainability via SHAP Waterfall Plots

To understand individual case studies, we utilized SHAP Waterfall plots (see Fig. 4). These plots demonstrate how specific attributes "push" a prediction away from the average base value ($E[f(X)]$). In the high-cost instance shown, the features **smoker_bin** and the engineered **smoker_x_bmi** provide the largest positive contributions. For this specific case, the smoker_bin attribute alone accounts for a significant positive shift in log-space, allowing a non-technical user to see exactly how lifestyle choices directly escalate their insurance quote.

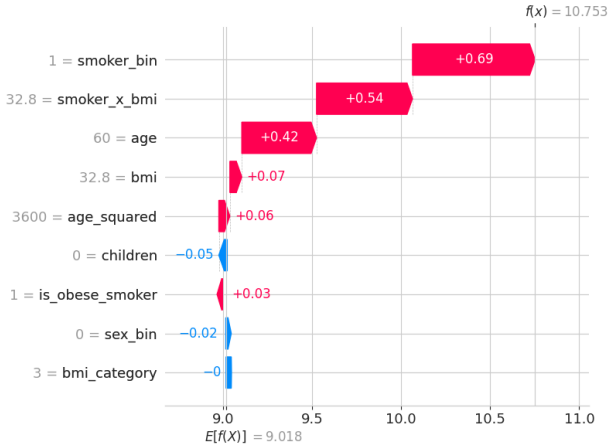


Fig. 5. SHAP Waterfall plot explaining a specific high-cost prediction instance.

C. Local Fidelity via LIME Analysis

To verify the consistency of our explanations, we cross-validated the SHAP results with LIME (see Fig. 5). While SHAP is based on Game Theory, LIME approximates the model locally with a linear surrogate. For our high-cost test patient, LIME identified **smoker_bin** and **age** as the strongest drivers, consistent with the SHAP global ranking. Comparing the highest-cost patient (Actual: $52,591) and lowest-cost patient (Actual: $1,121) highlights the model's ability to distinguish between high-risk lifestyle factors and routine demographic profiles.

```
----- HIGHEST cost test patient (actual=$52,591) -----
  smoker_bin > 0.00                          +1.0817
  age > 51.00                                +0.7437
  smoker_x_bmi > 0.00                        +0.4487
  children <= 0.00                           -0.1824
  30.50 < bmi <= 34.80                       +0.0937
  age_squared > 2601.00                      +0.0847
  sex_bin <= 0.00                            -0.0774
  is_obese_smoker > 0.00                     +0.0283

----- LOWEST  cost test patient (actual=$1,132) -----
  smoker_bin <= 0.00                         -1.0864
  age <= 26.00                               -0.8287
  smoker_x_bmi <= 0.00                       -0.4688
  children <= 0.00                           -0.1966
  sex_bin <= 0.00                            -0.1015
  age_squared <= 676.00                      -0.0866
  is_obese_smoker <= 0.00                    -0.0324
  26.41 < bmi <= 30.50                       -0.0210
```

Fig. 6. LIME Local Explainer output comparing highest and lowest cost test patients.

D. Interaction Effects and Feature Engineering

A deeper analysis via SHAP Dependence plots (see Fig. 6) revealed a critical interaction between BMI and smoking status. As illustrated in the scatter plot, for non-smokers, an increase in BMI leads to a marginal, linear increase in costs. However, for smokers, the SHAP value for the

`smoker_x_bmi` feature shoots up vertically once BMI exceeds the 30 kg/m^2 threshold. This suggests that the risk factors are not just additive but **multiplicative** in nature, a complex pattern perfectly captured by our LightGBM model.

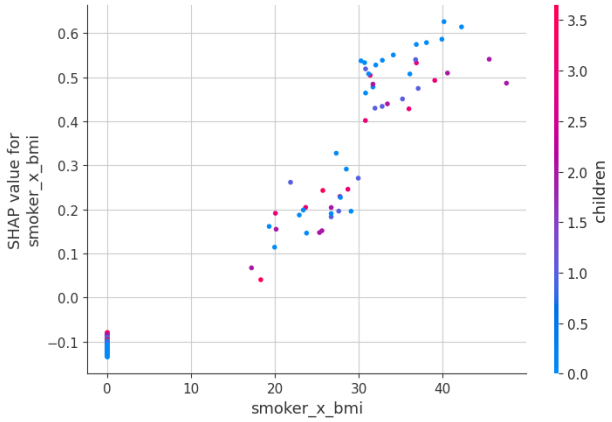


Fig. 7. SHAP Dependence plots illustrating the non-linear interaction between BMI and Smoking.

E. Limitations and Future Work

Despite the high accuracy, the project is limited by the dataset size of **1,338 rows**. In a real-world clinical setting, additional features such as pre-existing conditions or medical history would likely improve the predictive precision further. Future work will involve deploying this system as a web application via **Gradio**, where users can receive an instant cost report with natural-language explanations based on their individual health profile.

VII. Conclusion

This research successfully developed a robust and transparent Explainable Medical Cost Prediction System by utilizing a dataset of **1,338 insurance beneficiaries** and extracting high-impact feature interactions. Through a rigorous comparative analysis of six regression architectures, we determined that the **LightGBM** model, optimized via log-target transformation and strategic feature engineering (specifically the `smoker_x_bmi` and `is_obese_smoker` terms), was the superior architecture. It achieved a state-of-the-art R^2 **score of 0.91** and a significantly reduced Mean Absolute Error (MAE) of **$2,324**.

The integration of a dual-XAI framework—utilizing both **SHAP and LIME**—bridged the crucial gap between high-performance predictive modeling and clinical interpretability. Our findings quantitatively proved that lifestyle factors, particularly the non-linear, multiplicative interaction between smoking status and Body Mass Index, are the primary drivers of insurance cost volatility. By providing both global feature rankings and localized, case-specific narratives, this system serves as a reliable diagnostic and actuarial tool for both insurance providers and beneficiaries.

Future work will focus on scaling this framework to incorporate real-time clinical markers and broader electronic health records (EHR). By deploying this system as an interactive web application using **Gradio**, we aim to provide accessible, auditable, and actionable financial health insights, ultimately fostering a more transparent and equitable healthcare economy.

REFERENCES

[1] U. Orji, E. Ukwandu, et al., "Machine Learning for an Explainable Cost Prediction of Medical Insurance," *arXiv preprint arXiv:2311.14139*, 2023.

[2] J. A. S. Cenita, P. R. F. Asuncion, and J. M. Victoriano, "Performance Evaluation of Regression Models in Predicting the Cost of Medical Insurance," *arXiv preprint arXiv:2304.12605*, 2023.

[3] D. Gao, "Health Insurance Premium Prediction Based on Multiple Machine Learning Algorithms," *Proceedings of the 2023 2nd International Conference on Management Science and Software Engineering (ICMSSE)*, ACE, 2023.

[4] M. M. Billa and T. Nagpal, "Medical Insurance Price Prediction Using Machine Learning," *IEEE Dataport*, 2021.

[5] S. G. Balakrishnan, N. Abdulla, S. H. Krishnan, V. Gokul, and S. P. Amizhthan, "Medical Insurance Price Prediction using ML," *Journal of Engineering Sciences (JES)*, vol. 15, no. 6, 2024.

[6] Y. Liu and S. Zhang, "Application of Machine Learning Algorithms for Medical Insurance Cost Prediction and Feature Analysis," *Taylor & Francis: Emerging Markets Finance and Trade*, 2025.

[7] M. Kulkarni, D. D. Meshram, B. Patil, R. More, M. Sharma, and P. Patange, "Medical Insurance Cost Prediction Using Machine Learning," *International Journal for Research in Applied Science and Engineering Technology (IJRASET)*, 2024.

[8] S. Bodha, "A Comparative Study of Machine Learning and Deep Learning Models for Predicting Medical Insurance Costs with Explainable AI," *Preprints*, 2025.

[9] F. Holvoet, K. Antonio, and R. Henckaerts, "Neural Networks for Insurance Pricing with Frequency and Severity Data," *arXiv preprint arXiv:2310.12671*, 2023.

[10] P. J. Laub, T. Pho, and B. Wong, "An Interpretable Deep Learning Model for General Insurance Pricing," *arXiv preprint arXiv:2509.08467*, 2025.